

# Methadone, Metoclopramide and Metronidazole Interaction Causing Torsades de Pointes

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Dataset note: This case section was extracted mechanically from the cited PMC article. No clinical facts were added.

### 2. Case Report

A 50-year-old female with a medical history of acquired immunodeficiency syndrome not compliant with anti-retroviral therapy, polysubstance abuse and opioid dependence on methadone maintenance was brought in to the emergency department (ED) after she was found vomiting and confused in the streets. She reported using heroin and cocaine. On evaluation, she had sinus bradycardia, with a heart rate of 44 per minute, respiratory rate of 18 per minute and blood pressure of 128/80 mm Hg. Examination revealed a lethargic patient with no focal neurological deficits, clear lungs, sinus rhythm with no murmurs and soft abdomen. Laboratory testing showed a serum potassium of 3.5 meq/L, magnesium 2.1 meq/L, creatinine 0.5 mg/dL. Electrocardiogram (EKG) revealed sinus bradycardia with a rate of 40, a QT interval of 512 ms and a corrected QT (QTc) of 460 ms (Figure 1A). A non-contrast computerized tomography of the head and a chest radiograph did not show any acute abnormality. Her urine toxicology was positive for methadone, cocaine and opioids. Her mental status slowly improved during her stay in the ED; however, she started to have nausea and vomiting. She received intravenous (IV) metoclopramide 10 mg. She developed a fever of 101F during her stay in the ED. She was given IV ceftriaxone and metronidazole for suspected aspiration pneumonia. Metronidazole was started about 45 min after the metoclopramide was given and was slowly infused over 30 min. Shortly after the metronidazole infusion was stopped, she was unresponsive and the cardiac monitor (Figure 1B,C) revealed Torsades de Pointes (TdP). She was defibrillated with a return to sinus rhythm. Follow up EKG showed sinus rhythm with a QTc of 542 ms. Cardiology was consulted. IV potassium chloride and magnesium sulfate were given. Later, an echocardiogram showed an ejection fraction of 47%, normal right ventricular function and no valvular abnormalities. Over the next two days, there were no significant arrhythmias and the QTc returned to 432 ms.

#### Figure 1.

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- (A) Electrocardiogram on admission shows prolonged corrected QT (QTc) and variability in QT interval. (B) Onset of Torsades de Pointes (TdP) shortly after metronidazole infusion. (C) Persistent TdP requiring defibrillation.